

Afaf Abdelrahim

PhD Candidate

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EDUCATION

- Ph.D. Anticipated, Spring 2028 (GPA of 3.75), Ohio State University, Microbiology, and member of C-FARM, the Carbon Farming Alliance for Research and Management. Advisor: Virginia Rich.
 - *Thesis topic*: Interdisciplinary characterization of microbial roles in soil carbon stabilization under different agricultural practices.
- M.S. 2022 North Carolina State University, Microbiology. Advisor: Sid Thakur.
 - As a Fulbright Scholar, Fall 2020 – Spring 2022.
 - *Thesis title*: Phenotypic and Genotypic Characterization of Antimicrobial-Resistant *Escherichia coli* Isolated from Multiple Retail Meat Types in North Carolina.
- B.S. 2018 Modern Sciences and Arts University, Egypt, Pharmaceutical Sciences. Advisor: Reham Wasfi.
- *Honors thesis title*: Establishment of a PCR Based Diagnostic Method for The Identification Of *Klebsiella pneumoniae* In Clinical Specimens Targeting *rcaA* Gene In Egypt.

PUBLICATIONS *Also published under the surname Bastawy*

- **Abdelrahim A**, Harrell E, Fedorka-Cray PJ, Jacob M, and Thakur S. (2024). Phenotypic and genotypic characterizations of antimicrobial-resistant *Escherichia coli* isolates from diverse retail meat samples in North Carolina during 2018–2019. *Foodborne Pathogens and Disease*. 21(4).
- Ibrahim S and **Bastawy A**. (2020). Relevance of single-nucleotide polymorphism +62 G>A to the expression of resistin gene affecting serum resistin levels in metabolic syndrome in Egyptian population. *Current Pharmaceutical Biotechnology*. 21(7), 626-634.
- **Bastawy A** and ElRefaie H. (2019). Establishment Of a PCR Based Diagnostic Method for the Identification of *Klebsiella pneumoniae* In Clinical Specimens Targeting *rcaA* Gene in Egypt. *Global Journal for Research Analysis*. 8(3), 2277-8160.

Under review:

- **Abdelrahim A**, Demyan S, Lorenz K, Lal R, Rich VI. Effects of differing agricultural soil handling on microbial community characterization. Submitted to *Soil Biology and Biochemistry*.
- Mason L, Charles C, **Abdelrahim A**, Cronin D, Vik D, Li Y-F, Lorenz N, Sullivan MB, Rich VI, Dick R. "Microbial Mechanisms of Millet Drought Stress Mitigation in an Optimized Intercropping Shrub System". Submitted to *International Society of Microbial Ecology Journal*

In prep:

- **Abdelrahim A**, Mason L, Sun C, Vik D, Rich VI. Genome-resolved responses of the soil antibiotic resistome to shrub vegetation and drought. Target journal: *International Society of Microbial Ecology Journal*.
- **Abdelrahim A**, Mason L, Vik D, Rich VI. Metagenomic and Metatranscriptomic Insights into Microbial Carbon Storage Mechanisms Under Shrub Vegetation and Drought. Target journal: *Environmental Microbiology*
- **Abdelrahim A**, Ogg A, Lal R, Demyan S, Rich VI. Long-Term Tillage Management Shapes Soil Microbial Communities and Soil Carbon in Agricultural Soils. Target journal: *Soil and Tillage Research*
- **Abdelrahim A**, Ogg A, Lal R, Demyan S, Rich VI. Cross-Farm Variation in Bacterial, Fungal, and Arbuscular Mycorrhizal Fungal Communities in Agricultural Soils. Target journal: *Applied Soil Ecology*

TECHNICAL TRAINING

- College of Food and Agricultural Sciences “Using Generative and Agentic AI for Omics Data Analysis” workshop, summer 2026
- Translational Data Analytics Institute “AI in Your Research” workshop, summer 2026
- Marine Biological Laboratory (MBL), Woods Hole — Microbial Diversity Course, summer 2025: Intensive hands-on training in microbial isolation and cultivation, aseptic techniques, bright-field and fluorescence microscopy, and experimental microbiology.
- Environmental Molecular Sciences Laboratory (EMSL) Summer School, “Fundamentals and Complexities of Omics Data”, summer 2023
- NSF EMERGE Biology Integration Institute Summer Program, “Fundamentals of Interdisciplinary Biology and Team Science”, summer 2023
- Metabarcoding Workshop, Summer 2024
- Astrazeneca, Medical Communication Program, Summer Internship, summer 2017
- GlaxoSmithKline Microbial Quality Control Program, Summer Internship, summer 2016
- Pfizer Medical Communication Program, Summer Internship, summer 2014

SKILLS

- Field: Soil coring and sample processing by standard soil science methods, livestock (poultry) farm sampling by USDA-NIFA (United States Dept. of Agriculture - National Institute of Food and Agriculture) standardized protocols, environmental swab sampling, bulk density sampling, poultry litter and manure sampling
- Lab and molecular microbiology: Aseptic techniques, microbial isolation and cultivation, DNA extraction, PCR, gel electrophoresis, cloning, bright field and fluorescence microscopy
- Computational, programming and informatic tools: OSC High Performance Computing (HPC), vs code; python, R, bash/linux, Git/Github; QIIME2, PICRUST2, Lefse, DeePhage, iPHoP, Abriicate, CarveMe, Memote, CoBRAPy,
- Analytical: Metagenomic and metatranscriptomic data processing, genome-resolved metagenomics (MAGs), functional annotation, microbial community analysis, differential

abundance analysis, resistome analysis, ecological and co-occurrence network analysis, genome-scale metabolic modeling (FBA/GEMs), univariate and multivariate statistics, scientific data visualization, and reproducible bioinformatics workflows.

- Workflow: Git/GitHub, Nextflow (workflow development/management),
- AI:
 - Machine learning: random forest, XGBoost, scikit-learn, feature selection
 - AI-assisted bioinformatics and coding (Claude Code, GitHub Copilot) prompt engineering, and agentic AI workflows for scientific programming and pipeline development
- Team Science: scientific project management, grant proposal development, training and onboarding junior researchers, translating findings across disciplines, data and protocol standardization across collaborators, meeting facilitation, collaborative problem solving
- Languages: Arabic (fluent), English (fluent), German (basic)
- Music: Piano (level 2)
- Sports: Volleyball and Swimming

AWARDS

- Hidden Science Competition Awardee, Astera Institute, 2026. Awarded competitive funding for a research proposal selected by the Astera Institute to support publication and open dissemination of scientific findings.
- Microbial Diversity training Scholarship awardee, Marine Biological Laboratory, Woods Hole, 2025
- [EMSL Summer School Awardee](#), Pacific Northwest National Laboratory (PNNL), 2023- Competitively selected as one of 25 PhD students and postdoctoral researchers nationwide for advanced, hands-on training in multi-omics data analysis.
- Fulbright Scholar, 2020-2022
- First Place Undergraduate Thesis, judged by an external committee from Greenwich University (UK) and Zewail University (Egypt; founded by the Nobel Prize Chemistry Laureate Dr. Ahmed Zewail), of 400 student theses in the pharmaceutical sciences, 2018
- Awarded best Committee Chairperson by the Federation President for the Egyptian Pharmaceutical Student Federation (among 11 committees; I chaired the Professional Development Committee), 2017
- Student Merit Certificate Awardee, from the Faculty of Pharmacy, 2016
- Awarded best Committee Chairperson by the organization President for the Global Model World Health Organization (which I co-founded), 2016
- Pfizer Intern Talent Awardee, chosen as high-achieving intern and provided enrollment in *Pfizer Adopting Talent Program*, 2014

FUNDING

- Hidden Science Competition Award, Astera Institute, 2026 - \$10,000
- Cold Spring Harbor Laboratory Microbiome Meeting attendance scholarship, 2026 - \$850
- COMS Viromics training attendance scholarship, 2026 - \$500

- Woods Hole Marine Biological Laboratory partial attendance scholarship, 2025 - \$6,000
- NSF EMERGE Summer Program partial attendance scholarship, 2023 - \$3,000
- EMSL Summer School scholarship for on-campus attendance, 2023 - \$3,000
- Fulbright Scholarship, 09/20-06/22 - \$150,000

TEACHING EXPERIENCE

- Graduate Teaching Assistant, for “Introduction to Microbiology” (at Ohio State University, Dept. of Microbiology), Fall 2022, Spring 2023, Spring 2026
- Graduate Teaching Assistant, for “Microbial Genetics and Genomics” (at North Carolina State University, Dept. of Microbiology), Spring 2022
- Teaching Assistant, for “Biochemistry” (at Modern Sciences and Arts University, Egypt, Dentistry School, Dept. of Biochemistry), as post-baccalaureate, Fall 2018, Spring 2019, Summer 2020, Spring 2020, Summer 2020.

MENTORING EXPERIENCE

- Hannah Bernstein, undergraduate researcher from the Dept. of Food, Agricultural and Biological Engineering, Ohio State University, Fall 2023 - Fall 2024
 - *Undergraduate Honors Thesis*: “Analyzing how the agricultural microbiome impacts soil carbon stability under different management practices”
 - Denman Conference first place Awardee for poster presentation, March 2025, Columbus, OH
- Maizy Wilcox, undergraduate researcher from the Biology Program of Gallaudet University, 2025. This deaf student was my mentee via the EMERGE Biology Integration Institute REU program, Fall 2024 - Spring 2025
 - *Undergraduate Honors Thesis*: “Investigating the Microbial Communities in Freshwaters of Northern Sweden”
- Janelle Nimako, undergraduate researcher from the Dept. of Food, Agricultural and Biological Engineering, Spring 2025
- Nathan Favia, undergraduate researcher from the Dept. of Microbiology, Ohio State University, Spring 2024
- Sarah Kim, high school researcher from Dublin Coffman High School, Summer 2023

SYNERGISTIC ACTIVITIES

- Vice President for the Center of Microbiome Sciences Student Organization, Ohio State University, Jan 2025 - present
- Secretary for the Students for the Advancement of Microbiology, Jan-Dec 2024.
- Vice President of Fulbright Scholars Association at North Carolina State University, 2021
- Conference Moderator, National Institute of Antimicrobial Resistance Research and Education Conference, North Carolina State University, 2021
- Global Model World Health Organization (This student organization simulates the proceedings of the World Health Organization):
 - Co-Founder (2015)
 - Organizing Committee Head (2015-2016)
 - Graduate Advisor (2016-2017)

- Egyptian Pharmaceutical Student Federation (EPSF):
 - Professional Development Chairperson 2016-2017
 - Researcher at the research working group on Campus in 2015-2016
 - Diabetes Awareness Campaign conducted on the 6th of October city, Giza, Egypt in 2016.
 - Head of HCV Awareness Campaign that was conducted in the 6th of October city, Giza, Egypt in 2016.
 - Head of Breast Cancer Awareness Campaign that was conducted on the 6th of October city, Giza, Egypt in 2016.
 - Head of the Pharmacist Profession Development Project on campus in 2016.
- Pfizer: Organizer at *Read to Lead* in 2016.

PRESENTATIONS

- Invited seminar: "From who is there to what they do: decoding microbial communities across scales". Faculty of Pharmacy, Dept. of Microbiology, Modern Sciences and Arts University, Egypt, 2026.
- Poster presentation: "Agricultural tillage reshapes soil microbiomes across domains, including carbon stabilizing lineages" American Society for Microbiology ASM Microbe national conference, Washington, DC, 2026.
- Oral presentation: "Agricultural tillage alters soil microbiomes: across domains and including carbon stabilizing lineages" on the effects of agricultural tillage on bacterial and fungal communities using multi domain amplicon sequencing" [Soil Science Society of America, CANVAS national conference, Utah, 2025.
- Poster presentation: "How can harnessing microbiome-management interactions aid in climate change mitigation?". Byrd Center Polar and Climate Research conference, 2025
- Oral Presentation: "Soil microbiomes as allies in climate change mitigation". CFARM project workshop annual conference, Kansas, 2024.
- Oral presentation: "Phenotypic and genotypic characterization of antimicrobial resistant *E.coli* isolated from multiple retail meat in North Carolina" International Conference of Research Workers in Animal Diseases (CRWAD), Chicago, 2021.
- Oral presentation: "Whole genome sequencing based characterization of antimicrobial resistance and virulence genes in *Escherichia coli* isolated from retail meats in North Carolina (2018-2019)". National Institute of Antimicrobial Resistance Research and Education (NIAMRRE) annual conference, 2021.
- Oral presentation "Establishment of a PCR based diagnostic method for the identification of *Klebsiella pneumonia*". World Congress on Virology, Microbiology and Microbiologists in Orlando, 2018. Abstracts were published online.DOI: 10.4172/2324-8955-C2-009.
- Oral presentation "Establishment of a PCR based diagnostic method for the identification of *Klebsiella pneumonia*". Third International Conference on Diagnostic Microbiology and Infectious Diseases.Montreal Canada, 2018. Abstracts were published in the *Journal of Medical Microbiology and Diagnosis* 2018, Volume 7;DOI:10.4172/2161-0703-C2-021.